Depth and Shape Inference (I)

Introduction to Computational and Biological Vision

CS 202-1-5261

Computer Science Department, BGU

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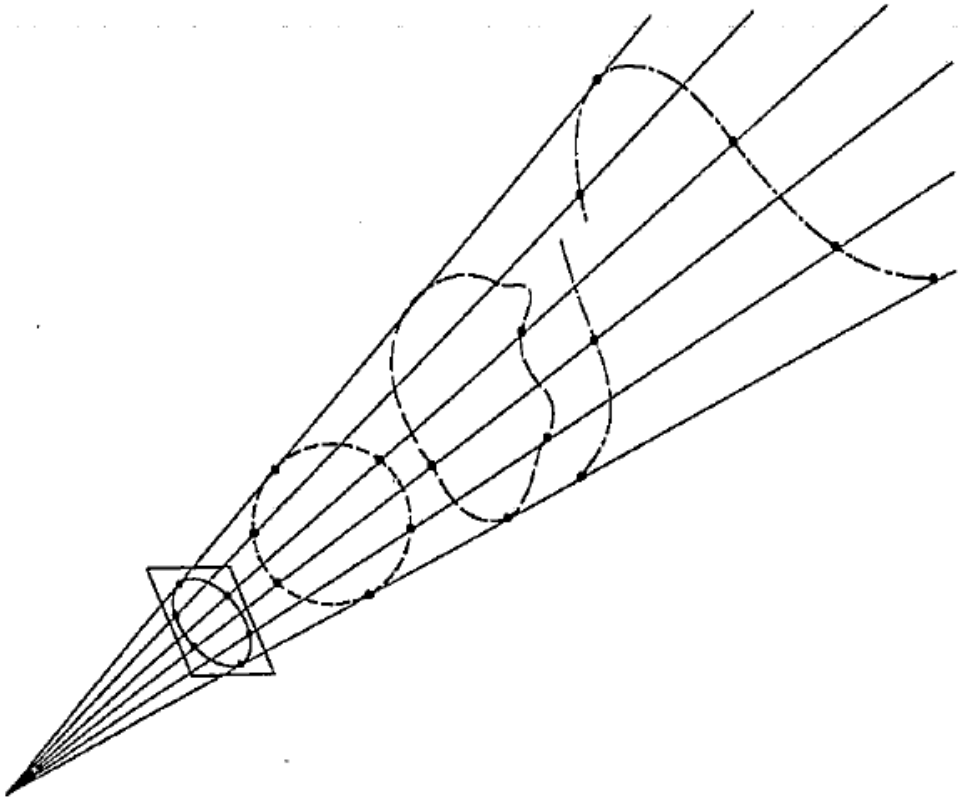
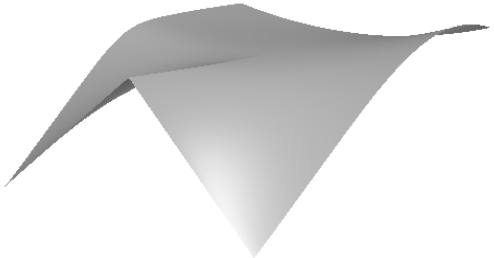
Depth and shape inference

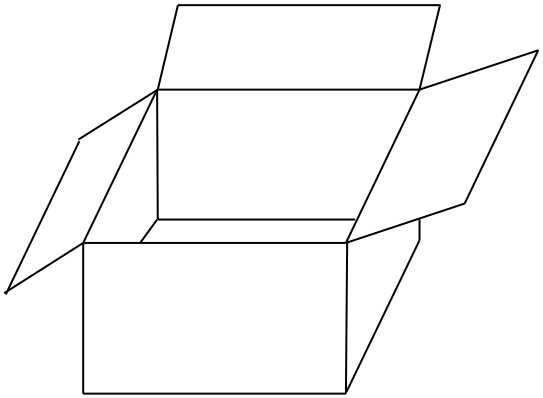
3D vision = inference of distance



Depth and shape inference

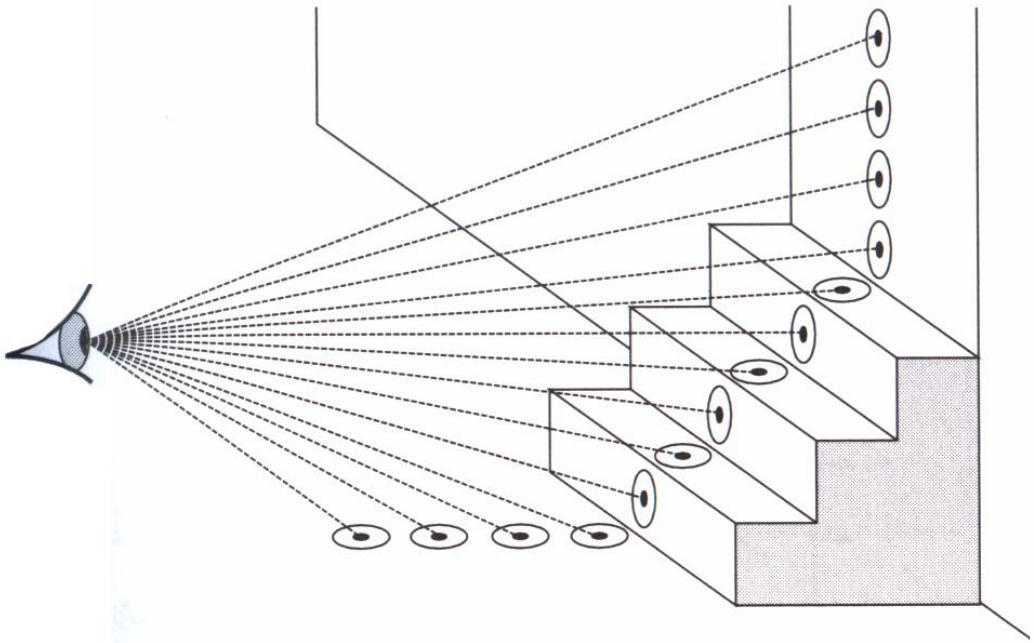
What does it mean to perceive "shape"?

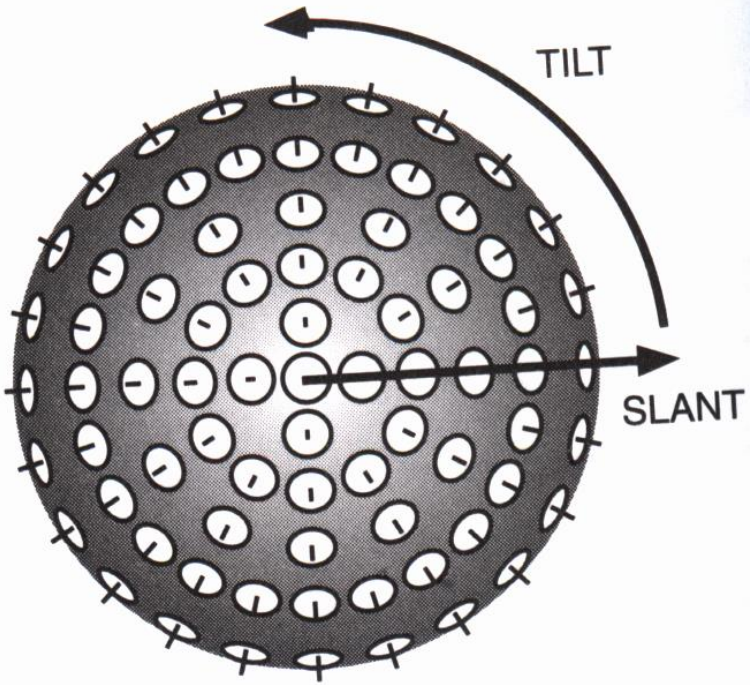




Depth and shape inference

Perception of surfaces oriented in space





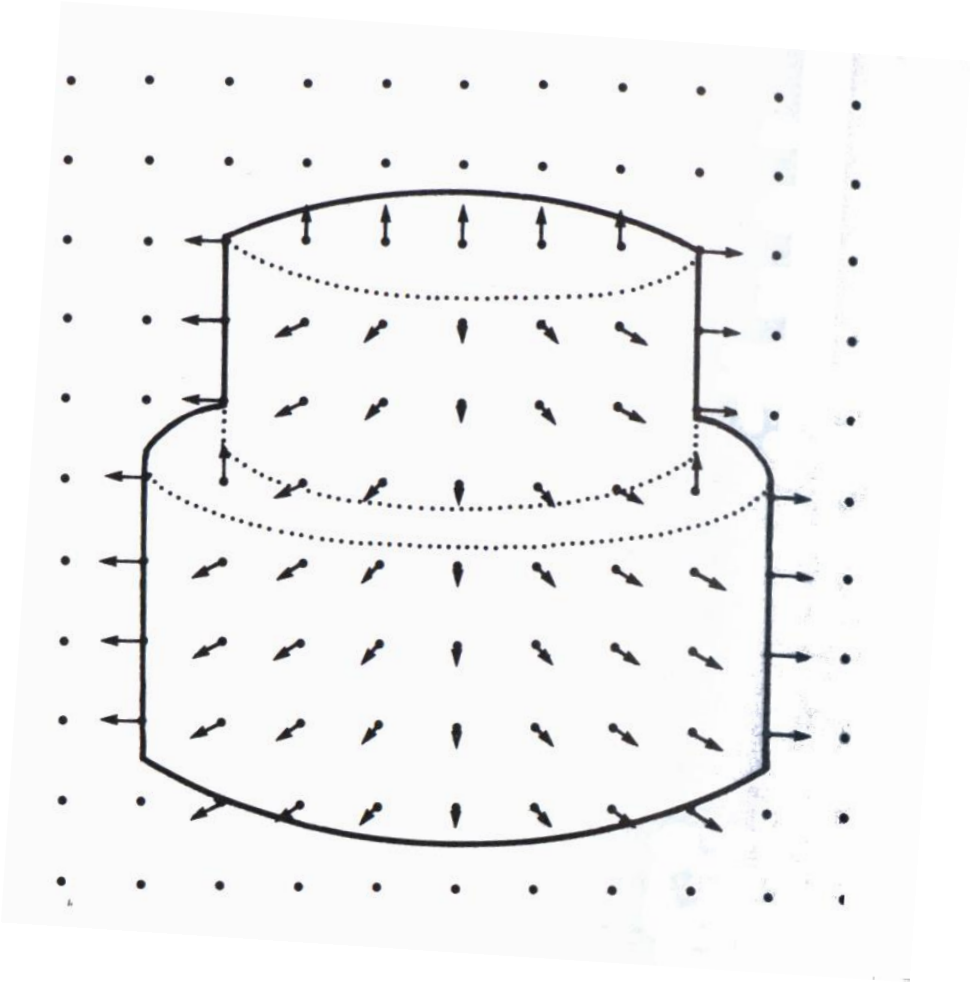
Depth and shape inference

Marr's 2.5D sketch

Image based representation
(Primal sketch)

• • •

Surface based representation
(2.5D sketch)



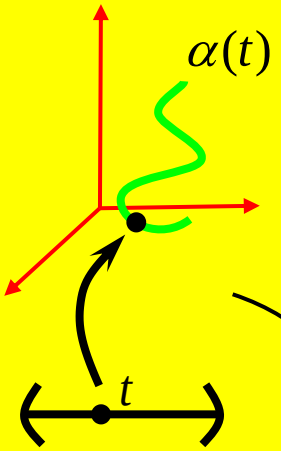
Depth and shape inference

Aside: What is a surface…

A parametrized differentiable 3D **curve** is a *differentiable* map

$$\alpha : (a,b) \to R^3$$

$$\alpha(t) = (x(t), y(t), z(t))$$



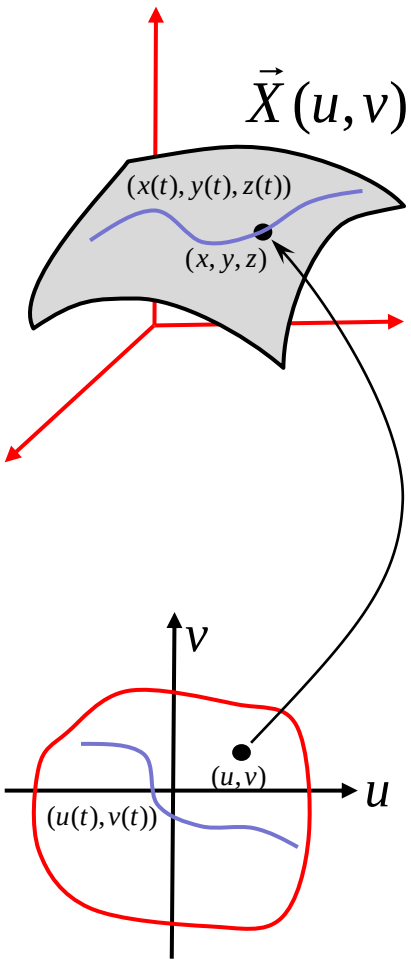
Depth and shape inference

Aside: What is a surface…

A parametrized differentiable **surface** in R^3 is a map

$$\vec{X}: U \subseteq R^2 \to R^3$$
$$\vec{X}(u,v) = (x(u,v), y(u,v), z(u,v))$$

whose coordinate functions ***x,y,z*** are differentiable.



<u>Observation</u>: A differentiable curve in the parameter space U maps to a differentiable curve **on the surface** in R^3

Depth and shape inference

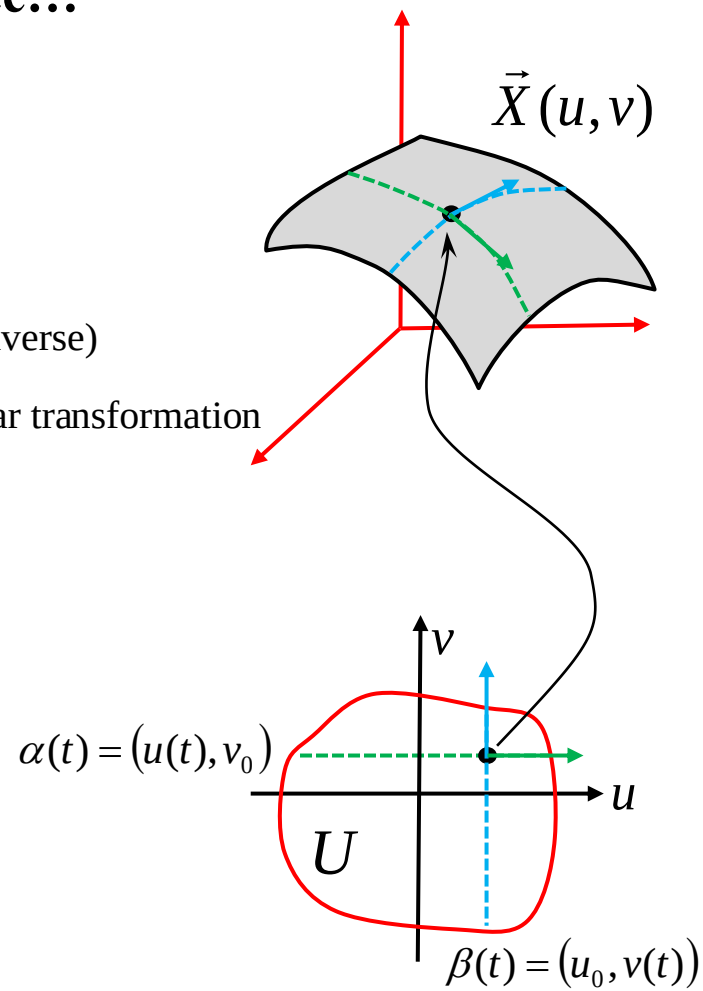
Aside: What is a surface…

A subset $S \subset R^3$ is a **regular surface** if for each $p \in S$ there exists a neighborhood $V \subset R^3$, an *open* set $U \subset R^2$, and a map $x: U \to V \cap S$ such that

1. $\vec{X}: U \to V \cap S$ is a parameterized differential surface.
2. $\vec{X}$ is a homeomorphism (continuous, one-to-one, onto, continuous inverse)
3. For each $q \in U$, the differential $d\vec{X}_q : R^2 \to R^3$ is a one-to-one linear transformation

   i.e., $\partial\vec{X}/\partial u$ and $\partial\vec{X}/\partial u$ are linearly independent.

$$d\vec{X}_q = \begin{bmatrix} \uparrow & \uparrow \\ \partial\vec{X}/\partial u & \partial\vec{X}/\partial v \\ \downarrow & \downarrow \end{bmatrix} = \begin{bmatrix} \partial x/\partial u & \partial x/\partial v \\ \partial y/\partial u & \partial y/\partial v \\ \partial z/\partial u & \partial z/\partial v \end{bmatrix}$$



Depth and shape inference

Aside: What is a surface…

Example: A **sphere** is the set

$$S^2 = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}$$

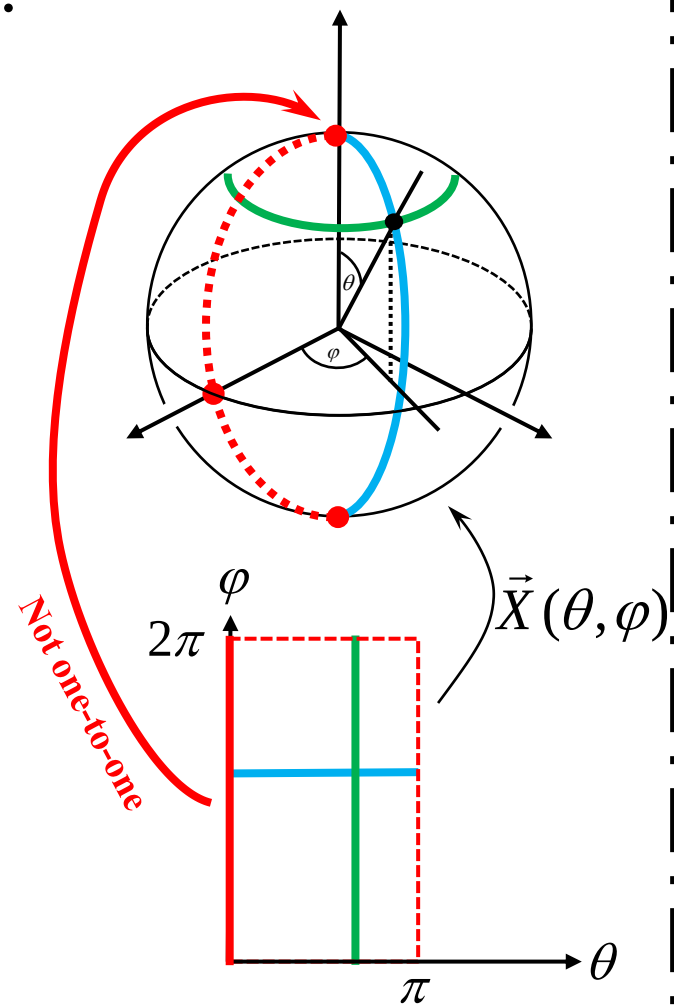
A sphere can be described as a parameterized surface using

$$\vec{X} = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$$

Is this parametrization *differentiable*?

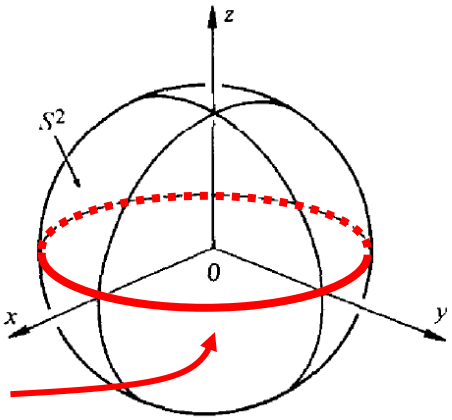
Is this parametrization really the entire sphere?

Is a sphere a *regular* surface?



Depth and shape inference

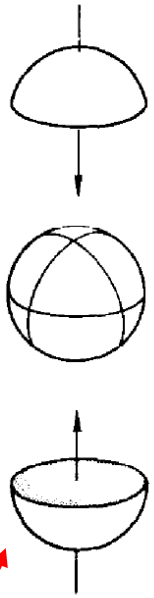
Aside: What is a surface…



$$\left.\begin{aligned}\mathbf{x}_1(x, y) &= (x, y, +\sqrt{1-(x^2+y^2)}),\\ \mathbf{x}_2(x, y) &= (x, y, -\sqrt{1-(x^2+y^2)}),\end{aligned}\right\}$$

Equator is missing

$$(x, y) \in U = \{(x, y) \in R^2;\, x^2 + y^2 < 1\}$$



$$\mathbf{x}_3(x, z) = (x, +\sqrt{1-(x^2+z^2)}, z),$$
$$\mathbf{x}_4(x, z) = (x, -\sqrt{1-(x^2+z^2)}, z),$$
$$\mathbf{x}_5(y, z) = (+\sqrt{1-(y^2+z^2)}, y, z),$$
$$\mathbf{x}_6(y, z) = (-\sqrt{1-(y^2+z^2)}, y, z),$$

A chart

Depth and shape inference

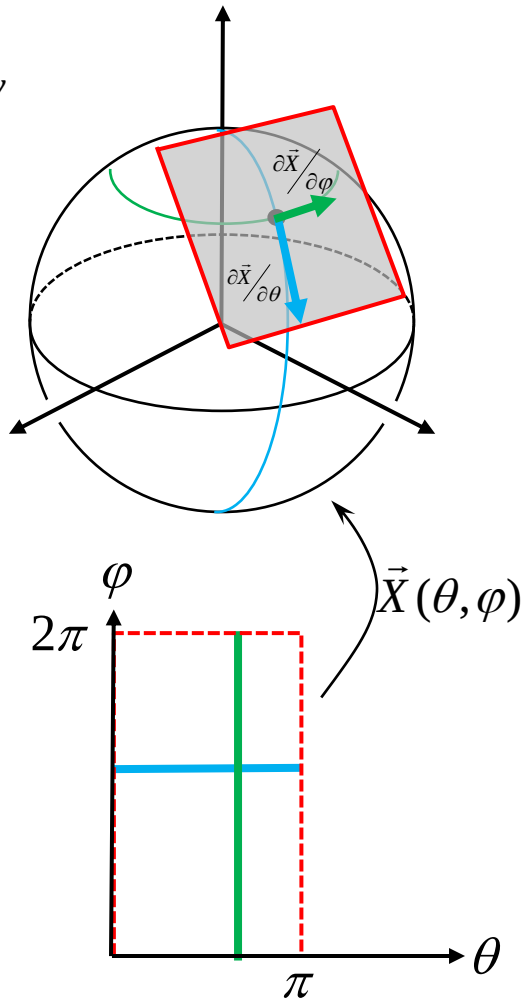
Aside: What is a surface…

The **tangent plane** of a differentiable surface is the plane spanned by $\partial\vec{X}/\partial u$ and $\partial\vec{X}/\partial v$

Example: for sphere we may obtain

$$\partial\vec{X}/\partial\theta = (-\sin\varphi\sin\theta, \cos\varphi\sin\theta, 0)$$

$$\partial\vec{X}/\partial\varphi = (\cos\varphi\cos\theta, \sin\varphi\cos\theta, -\sin\theta)$$

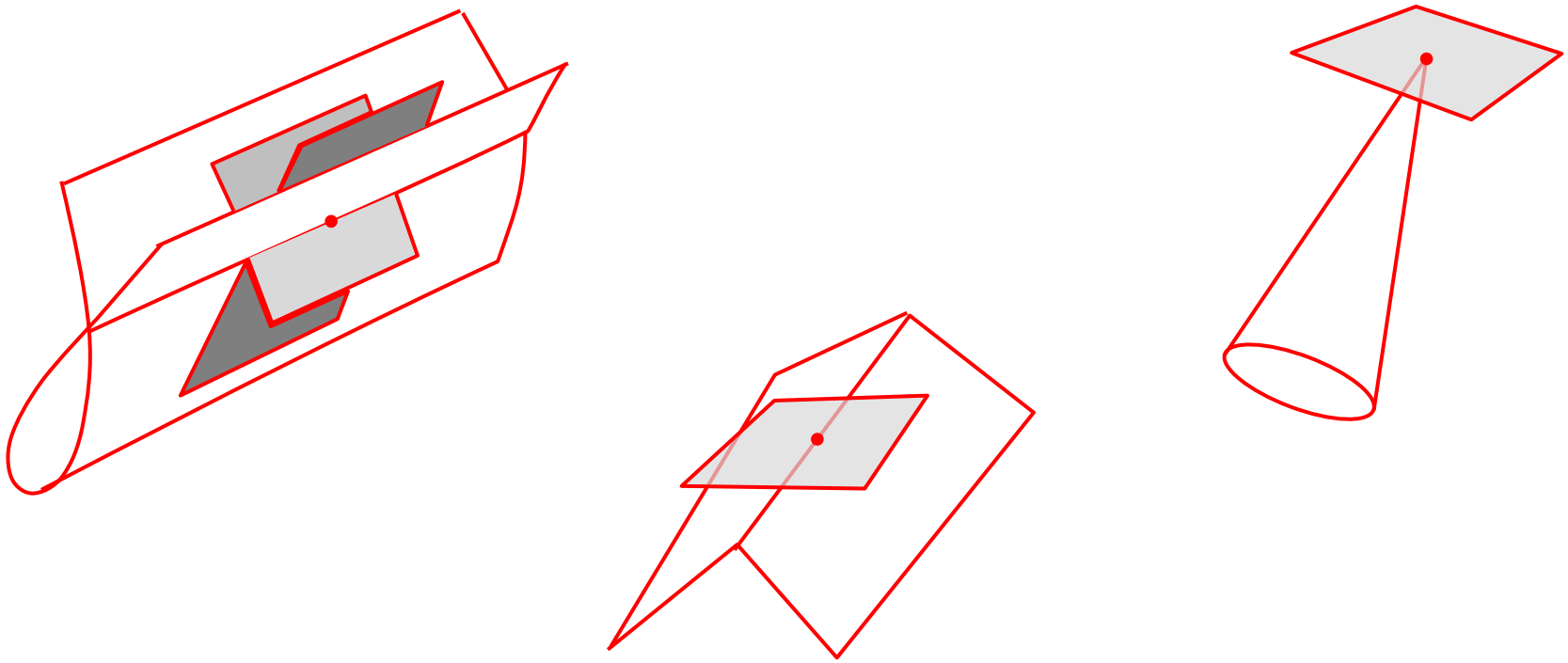


Observation:

A regular surface has a unique tangent plane at every point!

Depth and shape inference

Aside: What is a surface…



Examples of non regular surfaces

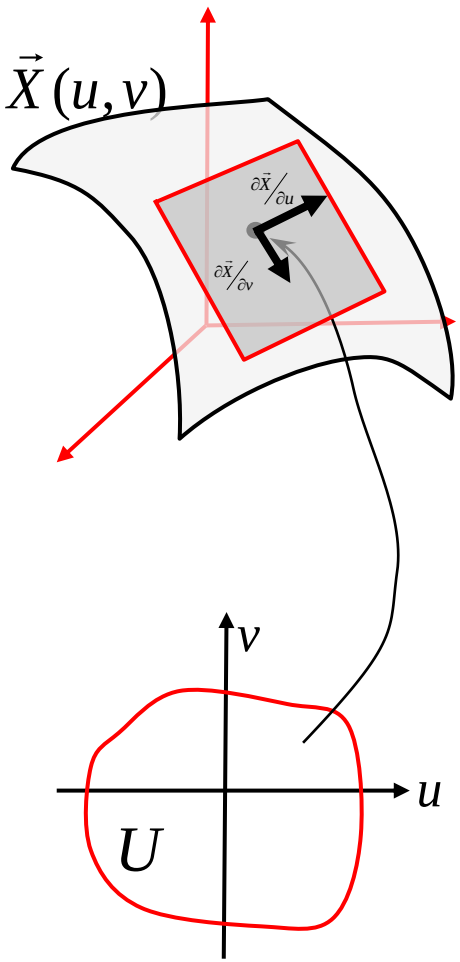
Depth and shape inference

Aside: What is a surface…

<u>A practical test for regularity</u>

$d\vec{X}_q$ is one-to-one iff $\partial\vec{X}/\partial u$ and $\partial\vec{X}/\partial u$ are linearly independent, thus

$$\boxed{d\vec{X}_q \text{ is one-to-one} \quad \Leftrightarrow \quad \partial\vec{X}/\partial u \times \partial\vec{X}/\partial v \neq \vec{0}}$$



Reminder:

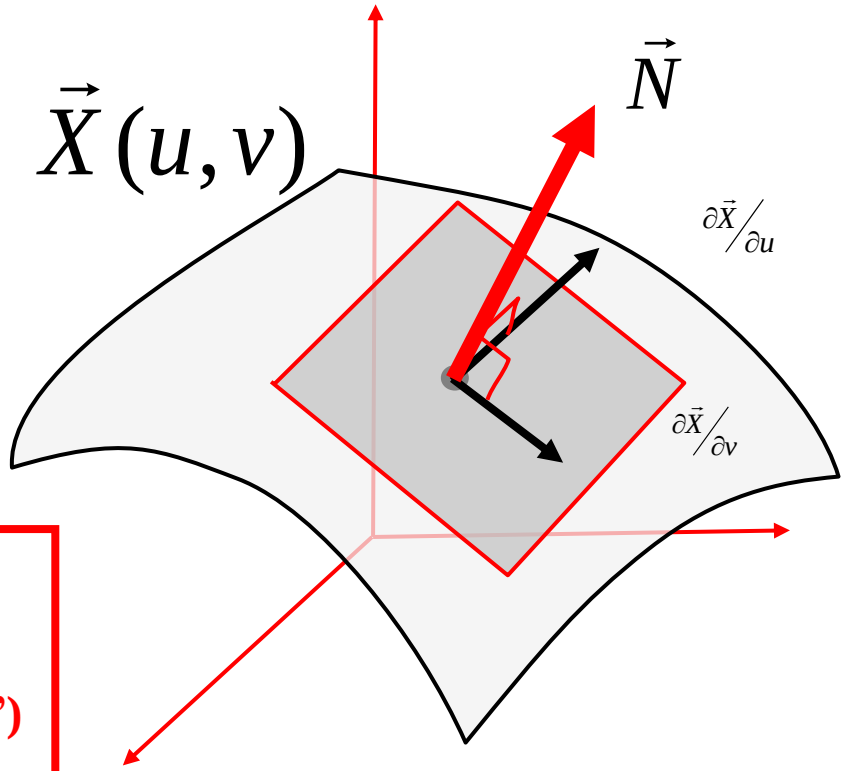
$$\partial\vec{X}/\partial u \times \partial\vec{X}/\partial v = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial x/\partial u & \partial y/\partial u & \partial z/\partial u \\ \partial x/\partial v & \partial y/\partial v & \partial z/\partial v \end{vmatrix} \neq \vec{0}$$

Depth and shape inference

Aside: What is a surface…

<u>The normal vercor of a regular surface</u>

$$\vec{N} = \frac{\partial\vec{X}/\partial u \times \partial\vec{X}/\partial v}{\left\|\partial\vec{X}/\partial u \times \partial\vec{X}/\partial v\right\|}$$

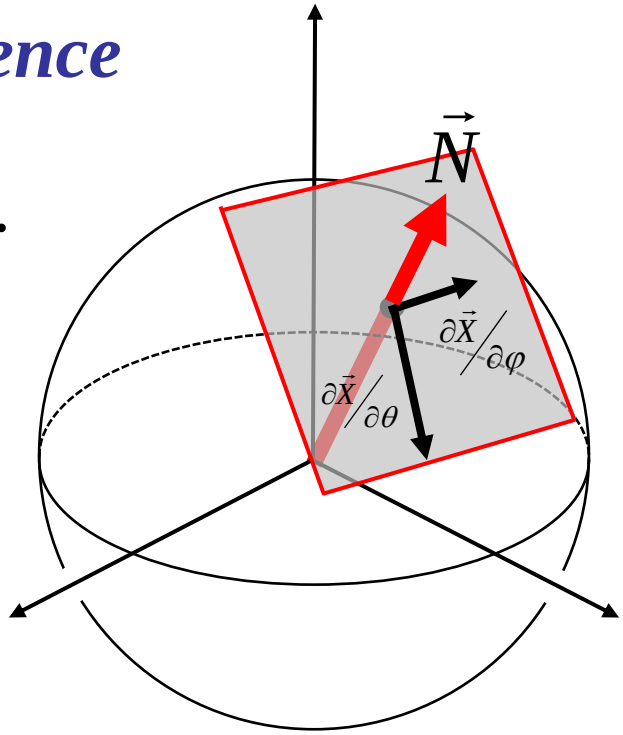


Corollary:

The regularity condition provides a local geometrical representation (and "direction") to the surface (through the tangent plane)

Depth and shape inference

Aside: What is a surface…



Example: for normal vector to a sphere

$$\vec{X} = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$$

$$\partial\vec{X}/\partial\theta = (-\sin\varphi\sin\theta, \cos\varphi\sin\theta, 0)$$

$$\partial\vec{X}/\partial\varphi = (\cos\varphi\cos\theta, \sin\varphi\cos\theta, -\sin\theta)$$

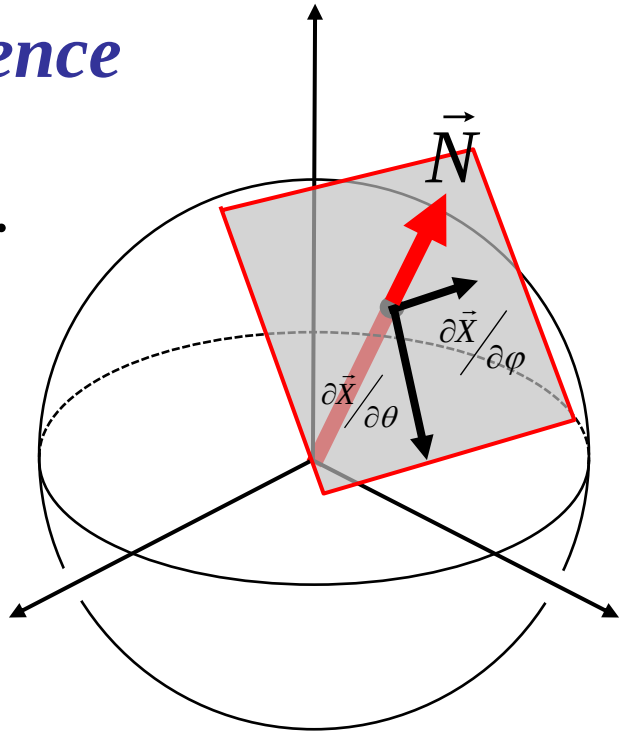
$$\partial\vec{X}/\partial\theta \times \partial\vec{X}/\partial\varphi = \left(\begin{vmatrix} \cos\varphi\sin\theta & 0 \\ \sin\varphi\cos\theta & -\sin\theta \end{vmatrix}, \begin{vmatrix} 0 & -\sin\varphi\sin\theta \\ -\sin\theta & \cos\varphi\cos\theta \end{vmatrix}, \begin{vmatrix} -\sin\varphi\sin\theta & \cos\varphi\sin\theta \\ \cos\varphi\cos\theta & \sin\theta\cos\varphi \end{vmatrix} \right)$$

$$\partial\vec{X}/\partial\theta \times \partial\vec{X}/\partial\varphi = (-\sin^2\theta\cos\varphi, -\sin^2\theta\sin\varphi, -\sin\theta\cos\theta) = -\sin\theta \cdot \vec{X}$$

$$\vec{N} = \frac{\partial\vec{X}/\partial\theta \times \partial\vec{X}/\partial\varphi}{\left\| \partial\vec{X}/\partial\theta \times \partial\vec{X}/\partial\varphi \right\|} = \vec{X}$$

Depth and shape inference

Aside: What is a surface…



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$$\vec{N} = \frac{\partial\vec{X}/\partial\theta \times \partial\vec{X}/\partial\varphi}{\left\| \partial\vec{X}/\partial\theta \times \partial\vec{X}/\partial\varphi \right\|} = \vec{X}$$

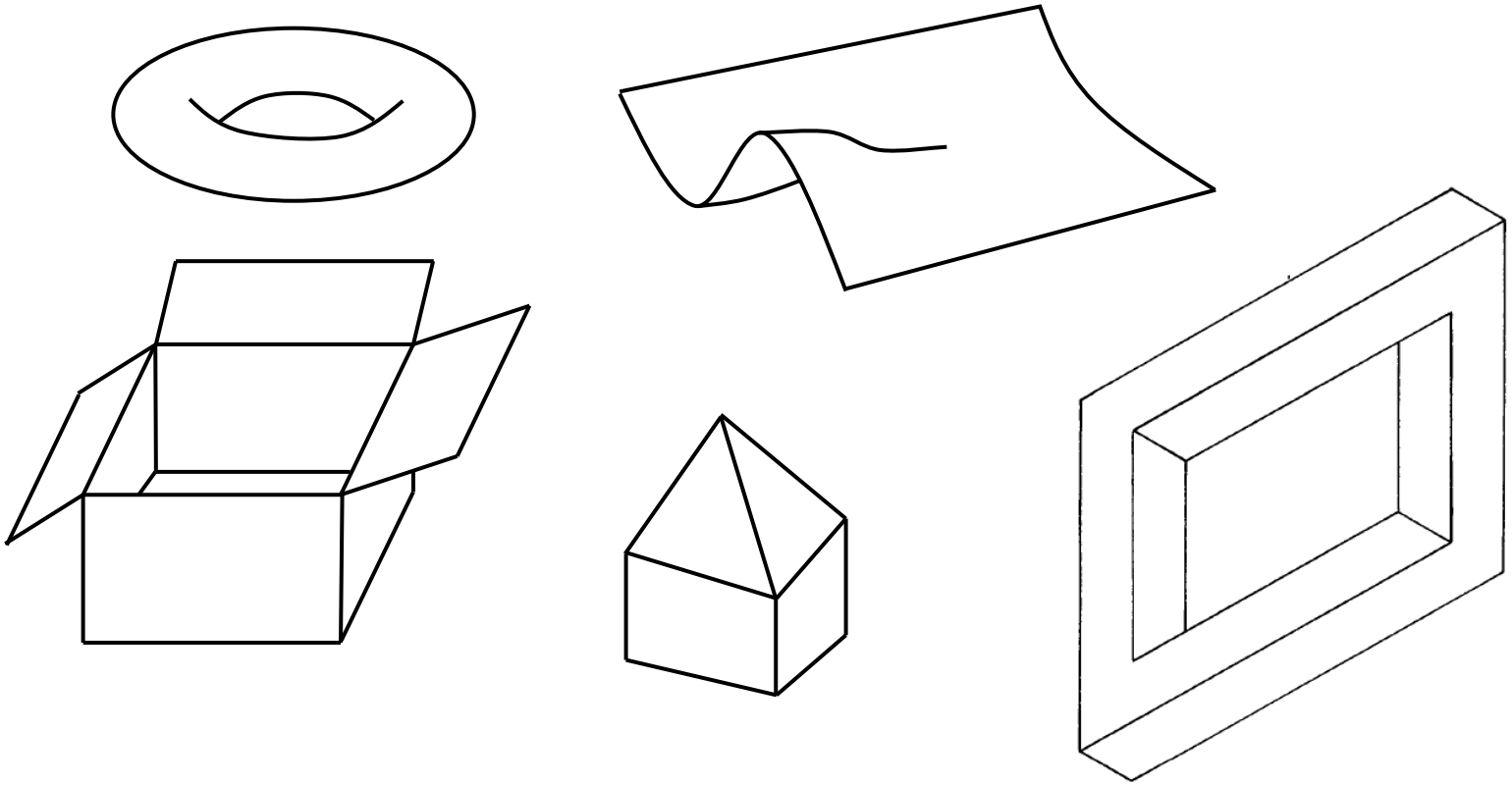
Depth and shape inference

Visual cues for depth and surface perception

Ocular cues :	Accommodation (monocular focus)
	Convergence (binocular fixation)
Pictorial cues:	Contours
	Shading
	Texture
	Shadows
	Convergence of parallel lines
	Relative size
	Aerial perspective
Stereoscopic information:	Binocular disparity
Dynamic cues:	Motion parallax
	Optical flow

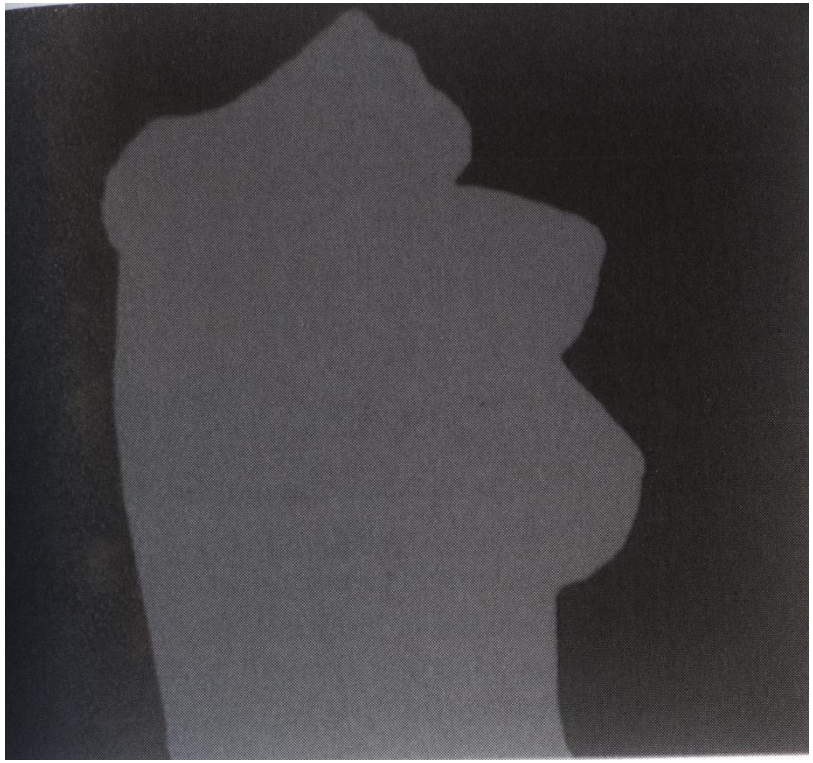
Depth and shape inference

Contours



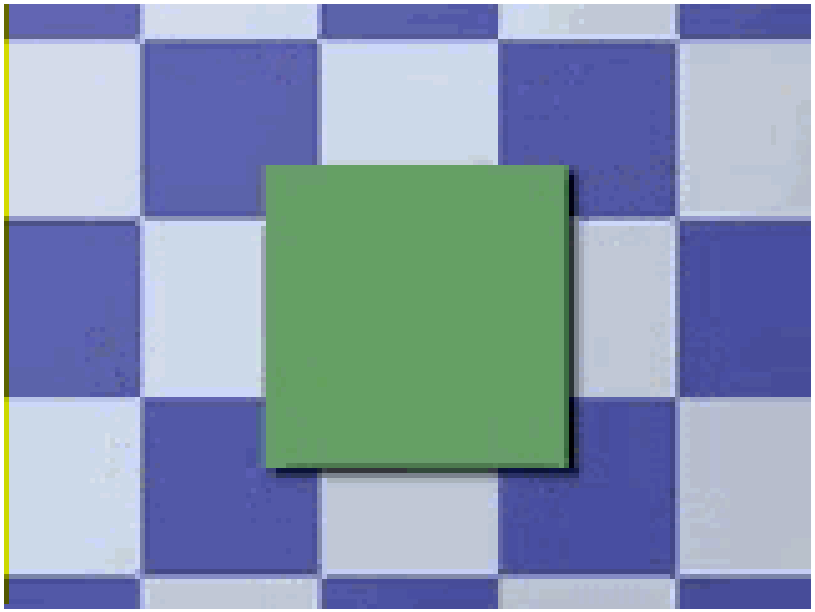
Depth and shape inference

Shading



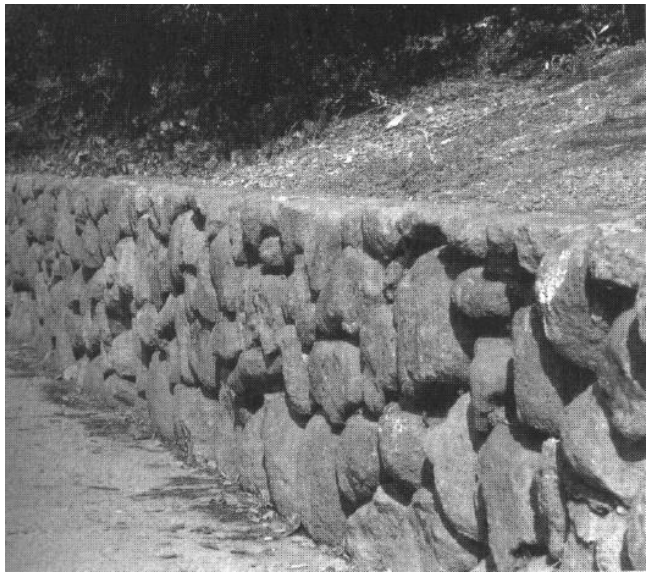
Depth and shape inference

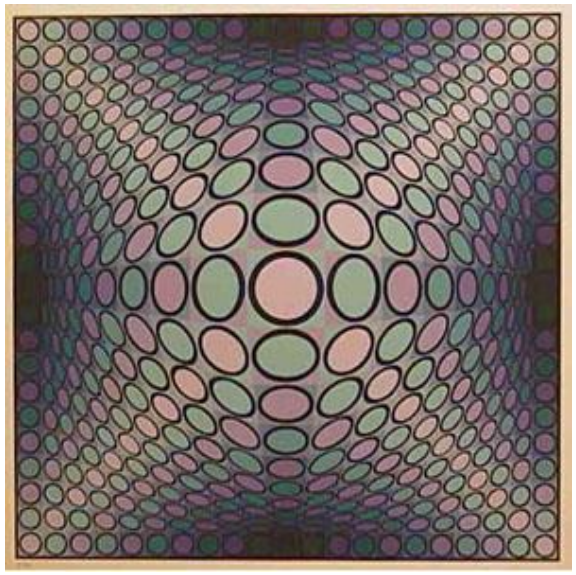
Shadows

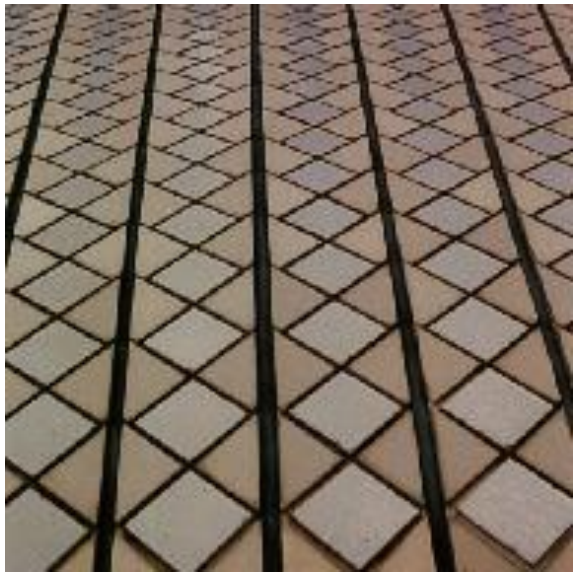


Depth and shape inference

Texture

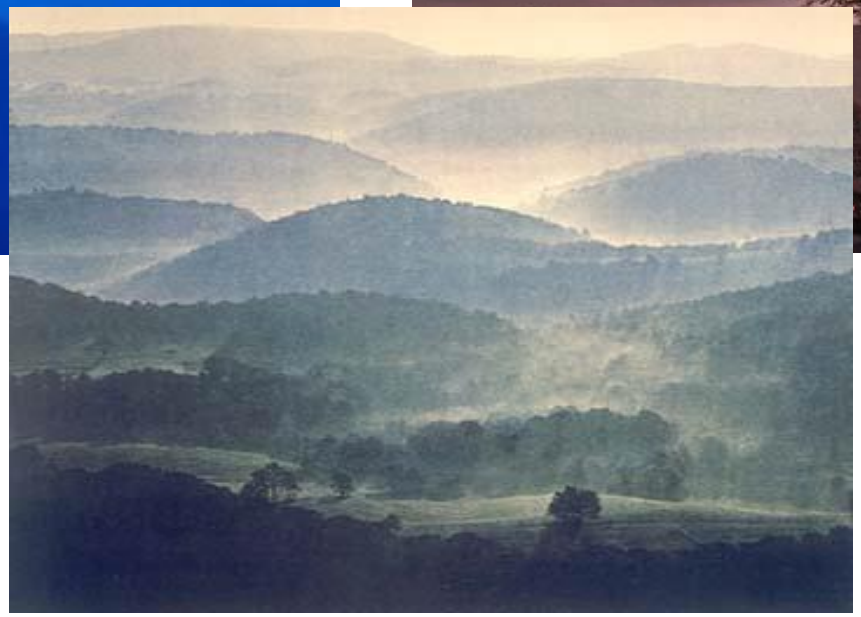






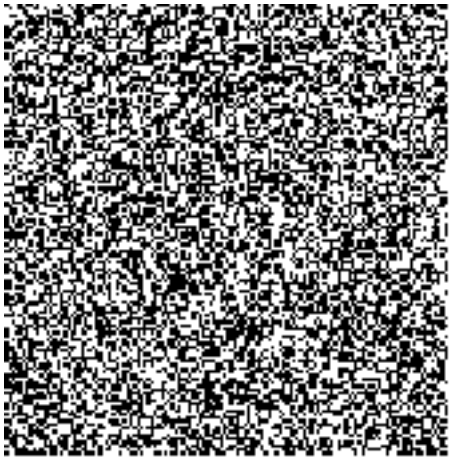
Depth and shape inference

Aerial perspective

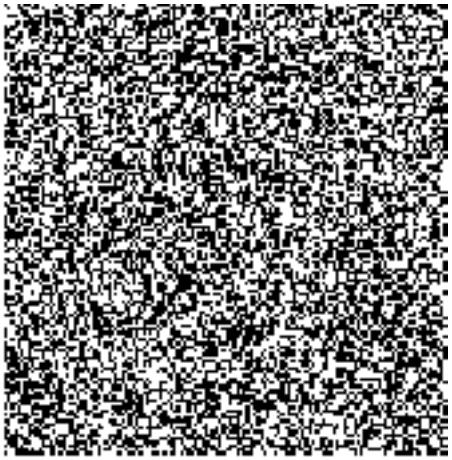


Depth and shape inference

Stereo



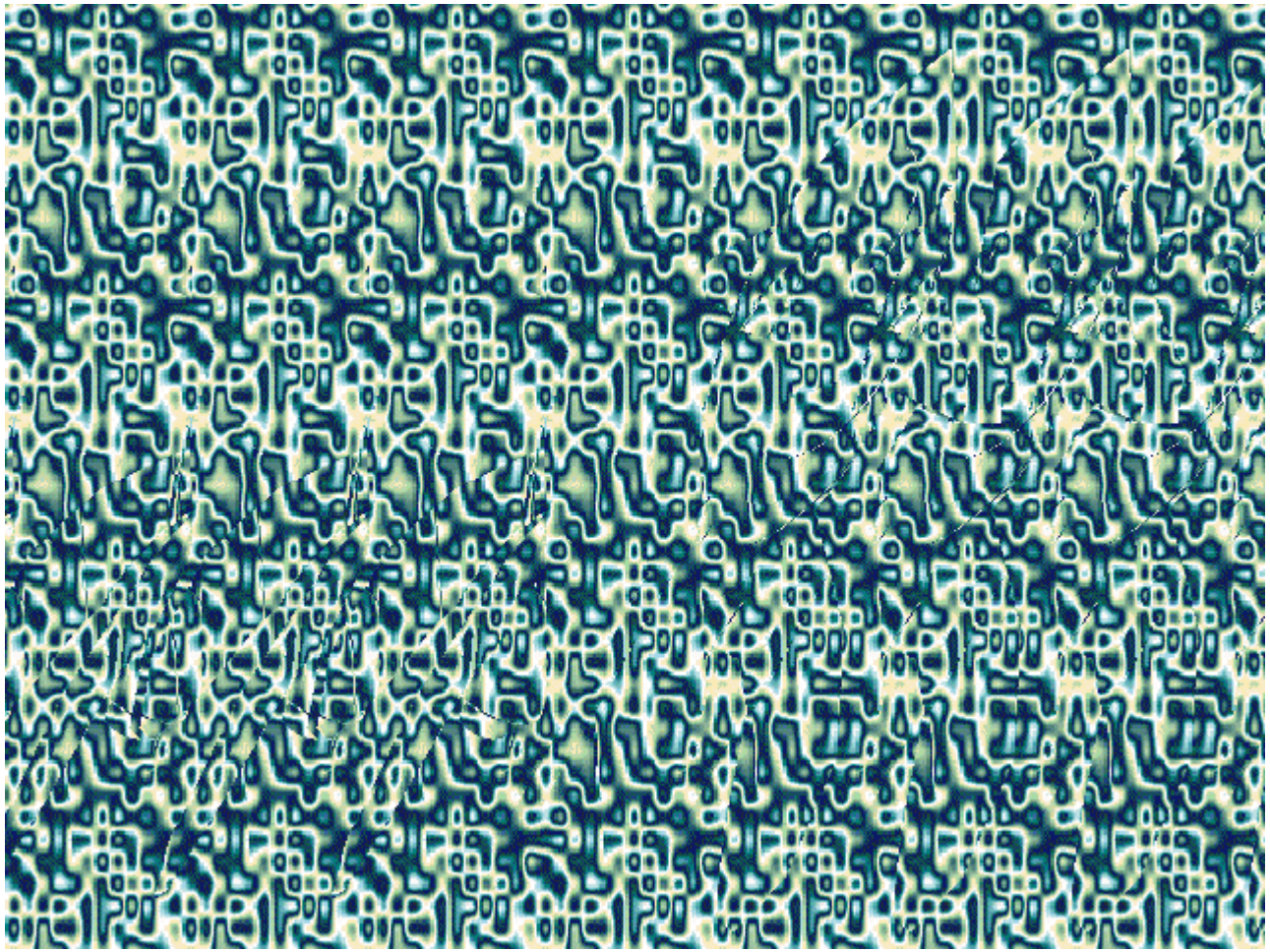
left

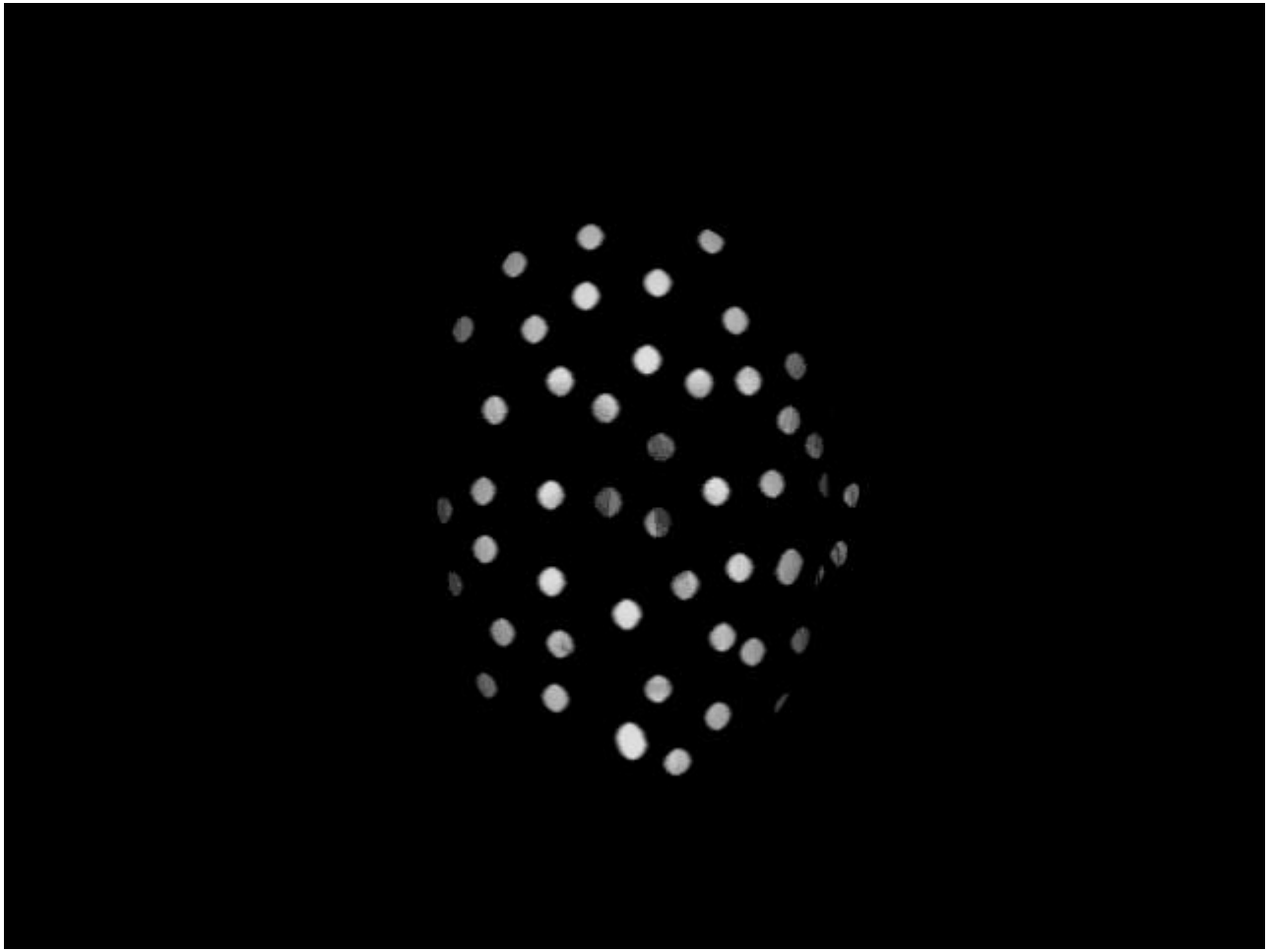


right

Depth and shape inference

Stereo



Depth and shape inference

Motion